CLOUD COMPUTING · 3-HOUR MODULE

Core Cloud Services

Compute · Networking · Storage · Messaging · Analytics

Google Cloud / with AWS & Azure equivalents

Lecture by **Imran Sayyed**

ROADMAP

Five Services We Will Cover

01

Compute

Running applications on virtual machines

02

Networking

Connecting and securing traffic

03

Storage

Keeping data durable and available

04

Messaging

Decoupling services with events

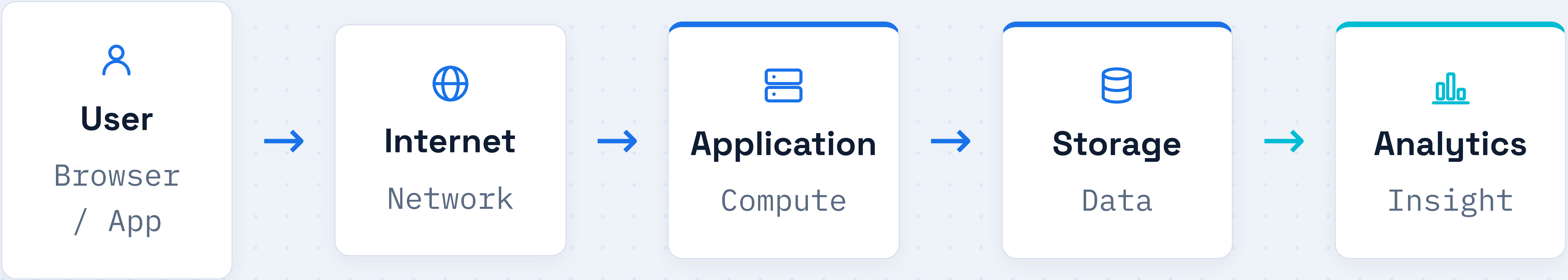
05

Analytics

Turning raw data into insight

FOUNDATIONS

How a Modern Application Works



Every cloud service in this course maps to one stage of this path.

FOUNDATIONS

Why Cloud Services Exist

The problems of running your own datacenter

01

High upfront cost

Buying servers, racks, and cooling before any users arrive

02

Slow to scale

New hardware takes weeks; demand spikes happen in minutes

03

Wasted capacity

You pay for peak load even when machines sit idle

04

Heavy maintenance

Power, repairs, security, and backups become your job

COMPUTE

What Is Compute?

The raw ingredients a machine needs to run software

PROCESS

CPU

Executes instructions

MEMORY

RAM

Holds active data

DISK

Storage

Keeps files persistently

LINK

Network

Connects to the world

COMPUTE

Introduction to Virtual Machines

Physical Server

One powerful machine of real hardware

↓

Hypervisor

Software layer that splits the hardware

↓

VM 1 | **VM 2** | **VM 3**

COMPUTE

Why Virtual Machines Exist

Better usage

One server runs many isolated machines instead of sitting idle

Fast to create

Launch or delete a VM in seconds, with no hardware to buy

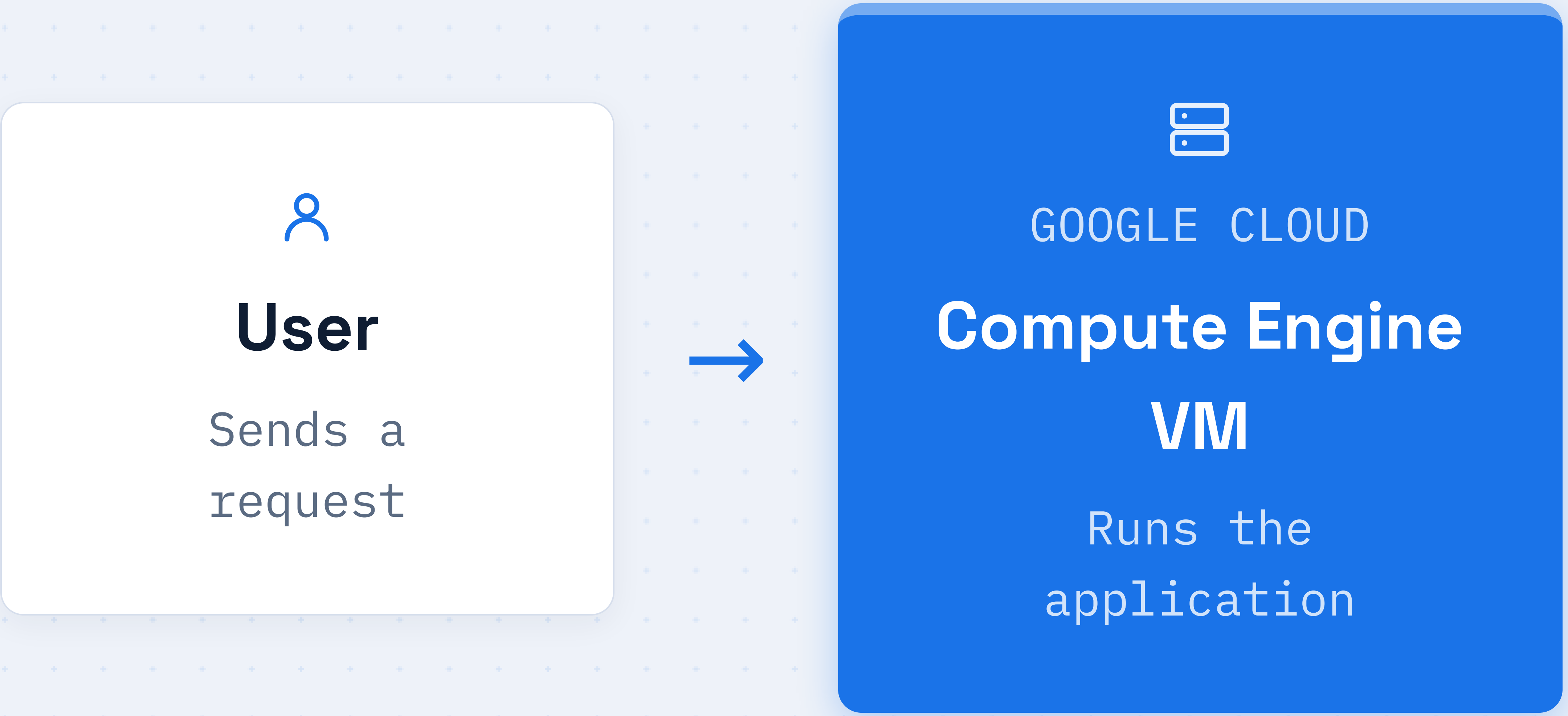
Isolation

Each VM is separate, so one workload cannot disturb another

COMPUTE · GOOGLE CLOUD

Compute Engine Overview

Google Cloud's service for renting virtual machines on demand



COMPUTE · GOOGLE CLOUD

Compute Engine Components

01
Machine Type
Preset size of the VM

02
CPU
Number of processing cores

03
RAM
Memory for active work

04
Disk
Persistent storage volume

05
Operating System
Linux or Windows image

06
IP Address
How the VM is reached

COMPUTE · AWS EQUIVALENT

Amazon EC2

GOOGLE CLOUD

Compute Engine

≈

AMAZON WEB SERVICES

Elastic Compute Cloud

EC2

EC2 instances are AWS virtual machines. You pick an instance type, an image, and storage, exactly as you would on Compute Engine.

COMPUTE · AZURE EQUIVALENT

Azure Virtual Machines

GOOGLE CLOUD

Compute Engine

≈

MICROSOFT AZURE

Azure Virtual Machines

Azure VM

Azure VMs offer the same model: choose a size, an OS image, and disks, then pay only while the machine runs.

COMPUTE

Compute Service Comparison

Concept	Google Cloud	AWS	Azure
Virtual machine	Compute Engine	EC2	Azure VM
Size unit	Machine type	Instance type	VM size
Pricing	Per second	Per second	Per second

COMPUTE · REAL WORLD

Running an E-Commerce Website

VMs scale up for a sale, then scale back down

Shoppers
Traffic spikes

→

Compute Engine
3 VMs → 30 VMs on sale day

→

Fast pages
No crashes

You only pay for the extra machines while the sale is busy, then release them afterward.

SECTION 02

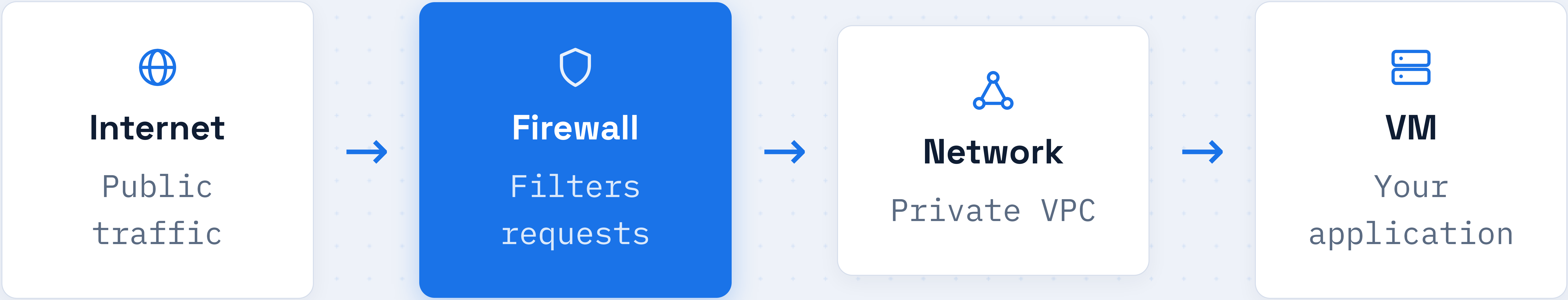
Networking

How traffic reaches your machines,
and how you keep it safe

02

NETWORKING

Cloud Networking Architecture



Traffic flows through a firewall and a private network before it ever touches your machine.

NETWORKING

Public IP

A reachable address on the open internet

Anyone online

From any country

→

PUBLIC IP

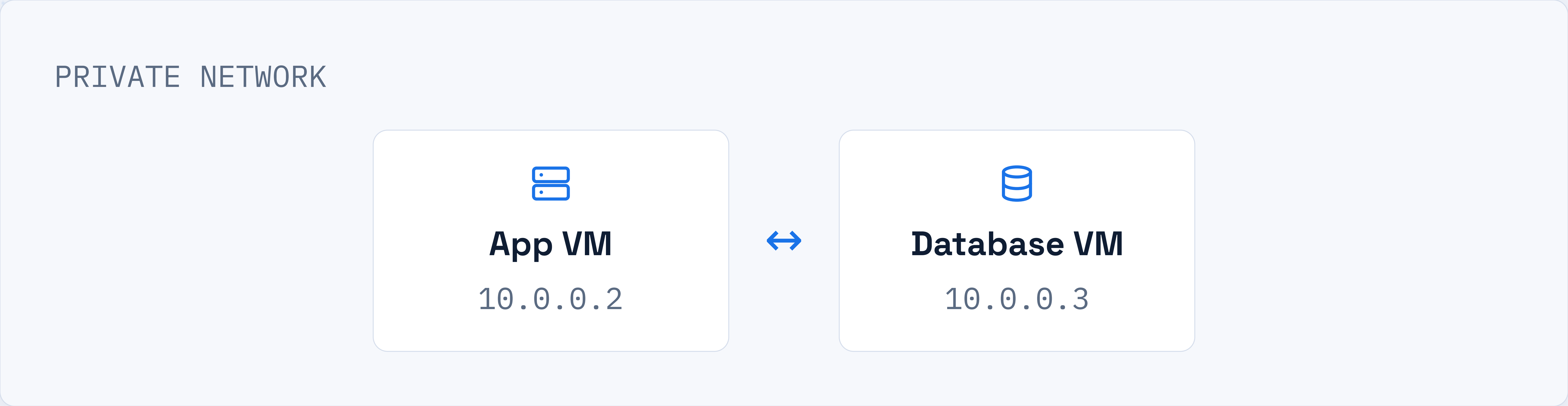
203.0.113.10

Web servers and load balancers use public IPs so users can find them. They are globally unique.

NETWORKING

Private IP

An address used only inside your own network

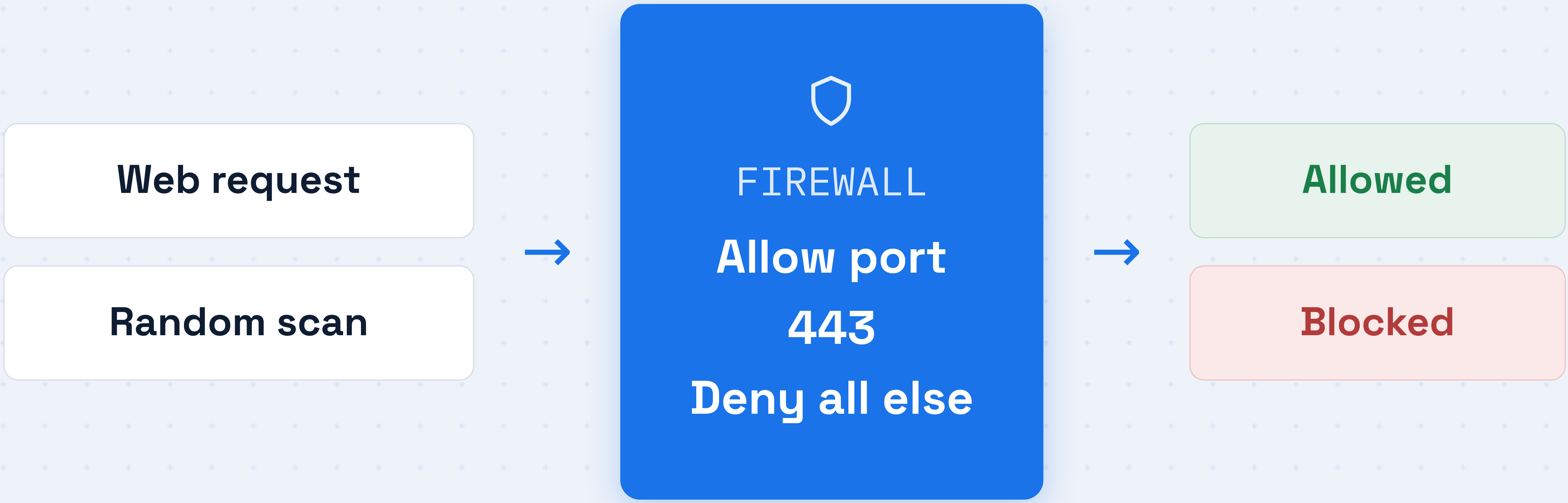


Private IPs let your machines talk to each other safely, hidden from the public internet.

NETWORKING

Firewall Rules

A gatekeeper that allows or blocks each request



NETWORKING · GOOGLE CLOUD

Virtual Private Cloud

Your own private, isolated network in the cloud

VPC NETWORK

SUBNET · WEB

Front-end VMs

SUBNET · DATA

Database VMs

A VPC groups your machines into subnets you fully control.

NETWORKING

Networking Service Comparison

Concept	Google Cloud	AWS	Azure
Private network	VPC	VPC	Virtual Network
Traffic filter	Firewall rules	Security groups	Network security groups
Traffic spread	Cloud Load Balancing	Elastic Load Balancing	Azure Load Balancer

SECTION 03

Storage

Where your data lives, and how you pick the right kind

STORAGE

Three Kinds of Storage

OBJECT

Object Storage

Files like photos, videos, and backups. Endlessly scalable.

BLOCK

Block Storage

A disk attached to one VM. Fast, like a hard drive.

FILE

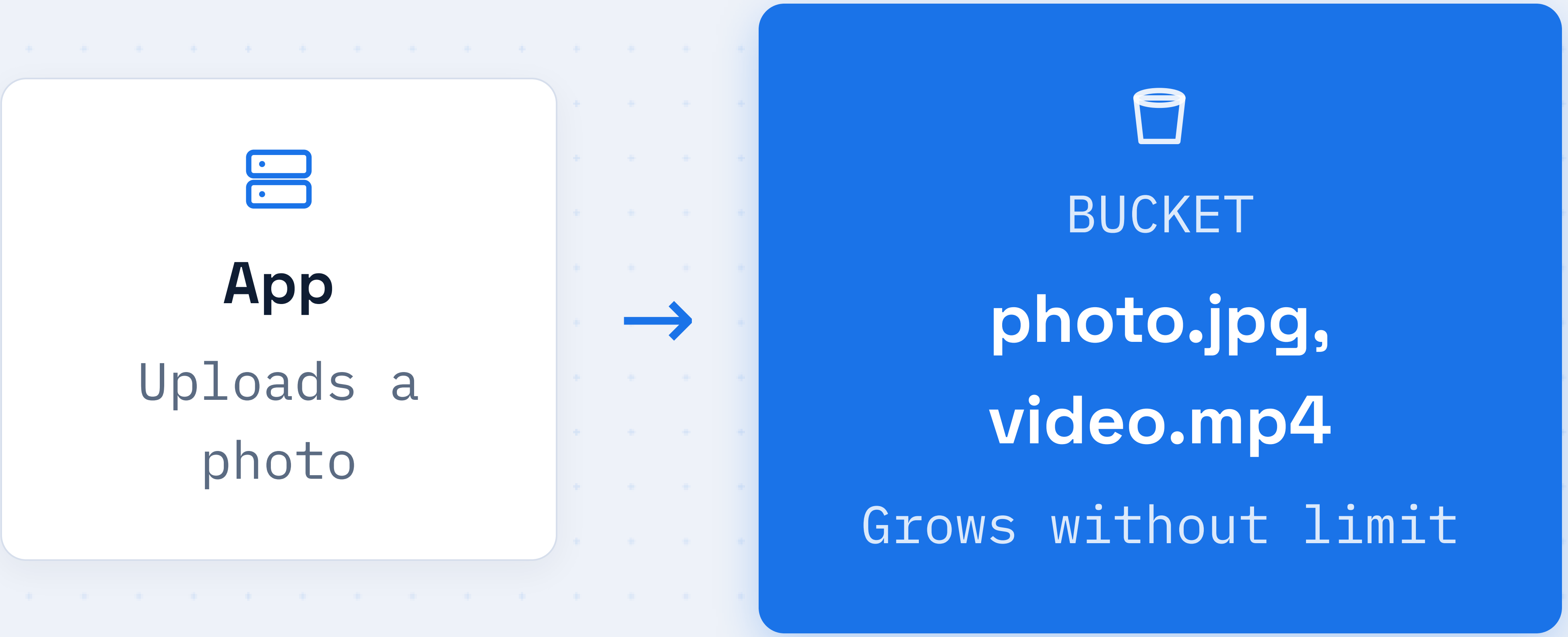
File Storage

A shared folder many VMs can read and write at once.

STORAGE · GOOGLE CLOUD

Object Storage

Google Cloud Storage holds files in buckets

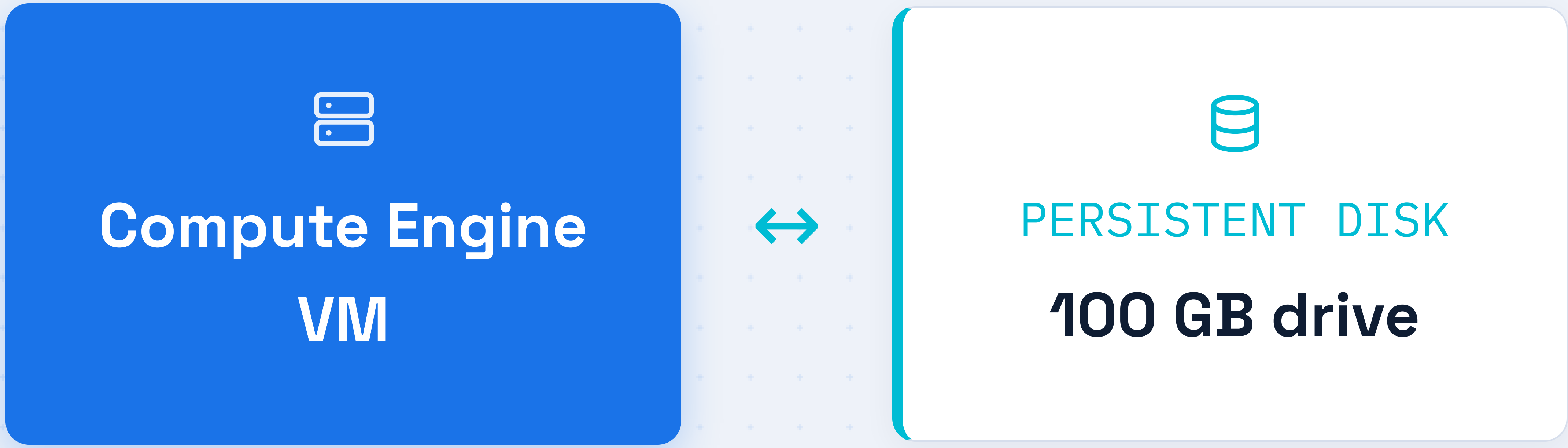


Each file gets its own web address, so apps and browsers can fetch it directly.

STORAGE · GOOGLE CLOUD

Block Storage

Persistent Disk attaches like a drive to one VM

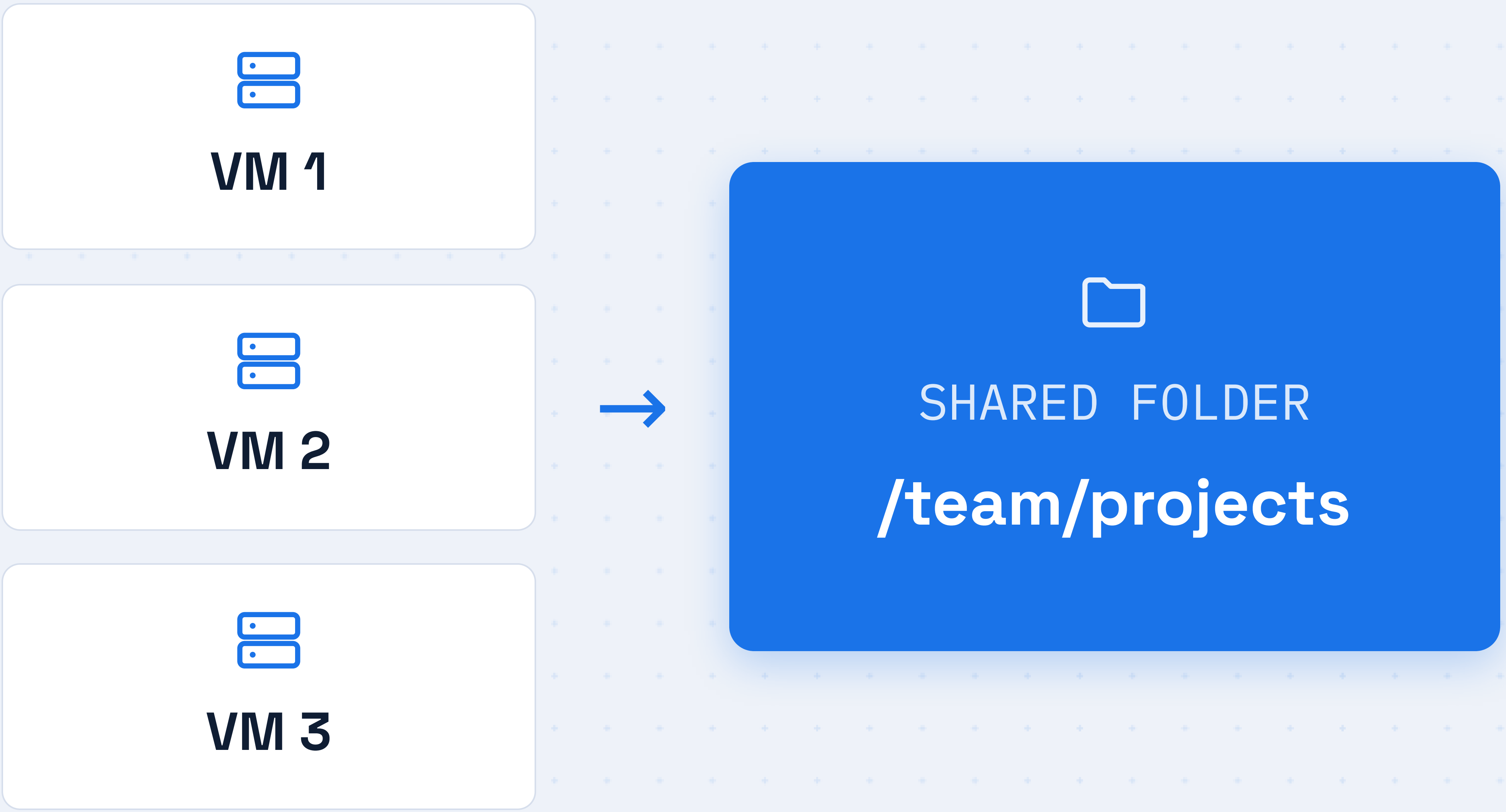


Block storage is fast and works well for databases and operating systems. The data stays even if the VM stops.

STORAGE · GOOGLE CLOUD

File Storage

Filestore is one shared folder for many VMs



Every machine sees the same files instantly, which is ideal for shared work.

STORAGE

Storage Classes

Pay less for data you rarely touch

HOT

Standard

Accessed often. Highest storage cost, instant access.

WARM

Nearline

Accessed monthly. Lower cost for backups.

COLD

Archive

Accessed yearly. Cheapest, for long-term keeping.

STORAGE · REAL WORLD

A Photo Sharing App

Each storage type does the job it is best at

OBJECT STORAGE

Uploaded photos

Millions of images served to users

BLOCK STORAGE

User database

Fast disk for accounts and captions

ARCHIVE CLASS

Old photos

Rarely viewed, stored cheaply

STORAGE · AWS EQUIVALENT

Amazon S3

GOOGLE CLOUD

Cloud Storage

≈

AMAZON WEB SERVICES

Simple Storage Service

S3

S3 also stores files in buckets and offers hot, infrequent, and glacier tiers, just like Cloud Storage classes.

STORAGE · AZURE EQUIVALENT

Azure Blob Storage

GOOGLE CLOUD

Cloud Storage

≈

MICROSOFT AZURE

Azure Blob Storage

Containers

Blob Storage keeps files in containers and offers hot, cool, and archive tiers that mirror the other clouds.

SECTION 04

Messaging

How services talk to each other without waiting around

04

MESSAGING

Why Messaging Exists

WITHOUT MESSAGING

Order service → Waits for email → Waits for billing

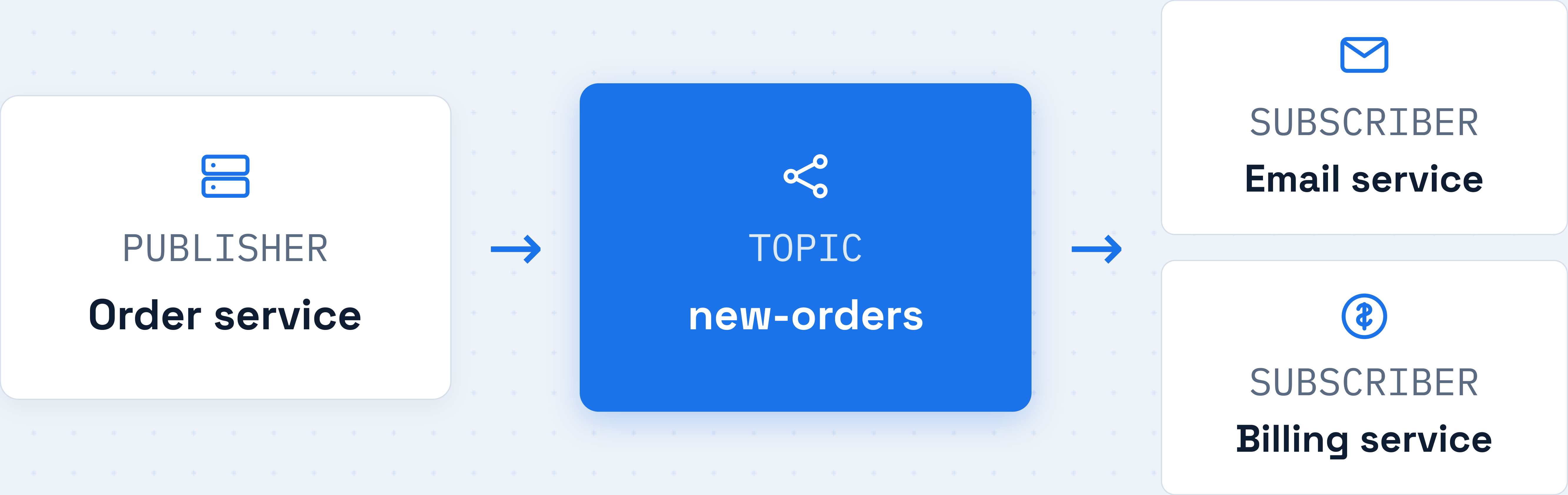
WITH MESSAGING

Order service → Drops a message → Moves on instantly

MESSAGING · GOOGLE CLOUD

Pub/Sub

Publishers send, subscribers receive, through a topic



One message can fan out to many subscribers at once.

MESSAGING

Two Ways to Deliver

PULL

Subscriber asks

The service checks the topic for new messages when it is ready to work.

Service → **Topic**

PUSH

Topic delivers

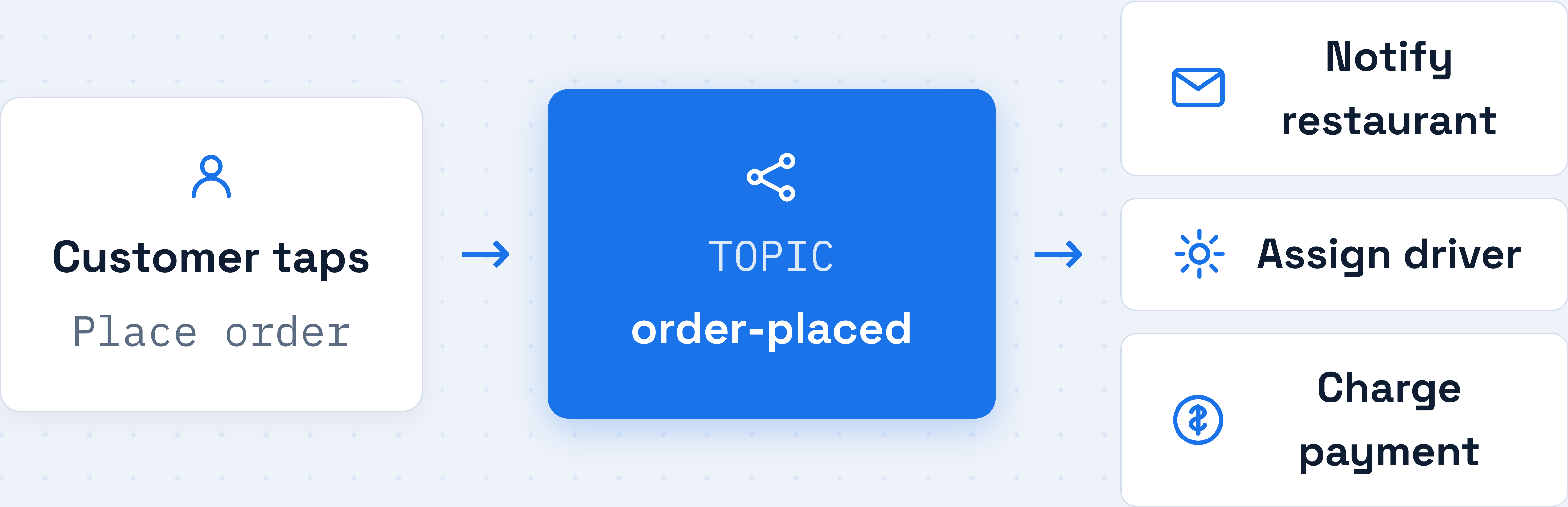
The topic sends each new message straight to the subscriber automatically.

Topic → **Service**

MESSAGING · REAL WORLD

A Food Delivery Order

One order message triggers many services

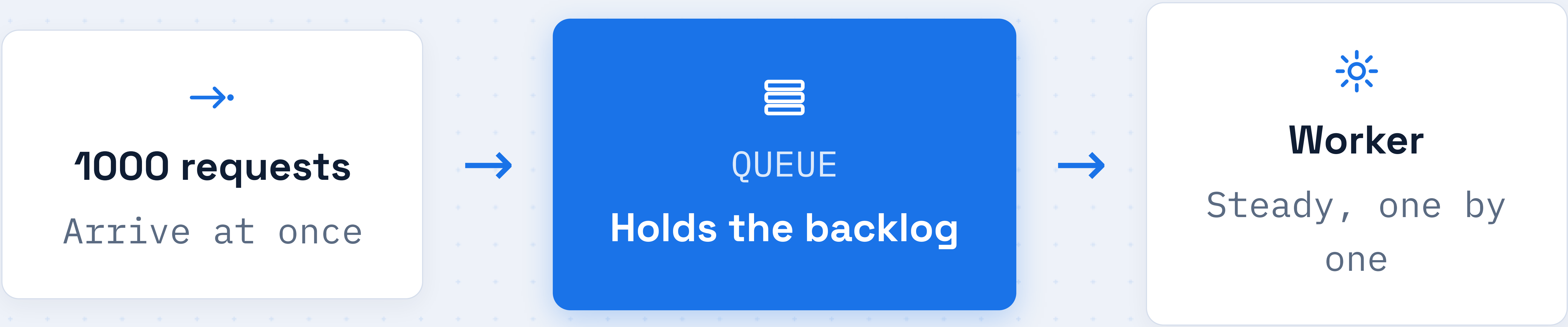


The app stays fast because it never waits for these jobs to finish.

MESSAGING

Queues Smooth Out Spikes

A queue holds work until a service can handle it



Nothing is lost during a rush, and the worker is never overwhelmed.

MESSAGING · AWS EQUIVALENT

Amazon SNS and SQS

GOOGLE CLOUD

Pub/Sub

≈

SNS · NOTIFICATIONS

Simple Notification Service

SQS · QUEUES

Simple Queue Service

AWS splits the job in two: SNS fans out messages, SQS holds them in a queue.

MESSAGING · AZURE EQUIVALENT

Azure Service Bus

GOOGLE CLOUD

Pub/Sub

≈

MICROSOFT AZURE

Azure Service Bus

Topics and queues

Service Bus offers both topics and queues in one service, covering the same patterns as Pub/Sub.

SECTION 05

Analytics

Turning huge piles of data into answers you can act on

05

ANALYTICS

What Analytics Solves

A normal database struggles with billions of rows

REGULAR DATABASE

Built for one record at a time

Great for "find this order", slow at "sum 5 years of sales".

ANALYTICS WAREHOUSE

Built for huge questions

Scans billions of rows in seconds to find trends.

ANALYTICS · GOOGLE CLOUD

BigQuery

Ask questions in plain SQL, get answers fast

DATA

Billions of rows

→

BIGQUERY · SQL

SELECT … GROUP BY

→

INSIGHTS

Charts and trends

You do not manage any servers. BigQuery spreads the work across thousands of machines for you.

ANALYTICS · GOOGLE CLOUD

How BigQuery Is So Fast

SPLIT WORK

Thousands of workers

A query is divided and run in parallel.

SEPARATE

Storage from compute

Each scales on its own, so neither holds the other back.

PAY PER QUERY

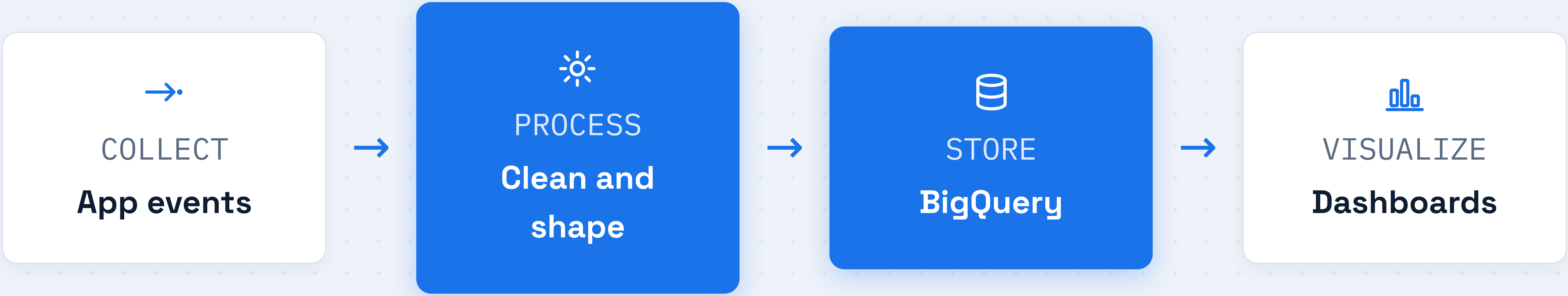
Only for data scanned

No idle servers to pay for between questions.

ANALYTICS

A Data Pipeline

Raw events become clean, query-ready data



Each stage has a job, and the data flows steadily from left to right.

ANALYTICS · REAL WORLD

A Streaming Service

Viewing data guides what to recommend next

Millions of plays

Logged every day

BIGQUERY

Find top shows

Better picks

For each viewer

Insights from analytics feed straight back into the product the user sees.

ANALYTICS · AWS EQUIVALENT

Amazon Redshift

GOOGLE CLOUD

BigQuery

≈

AMAZON WEB SERVICES

Amazon Redshift

Data warehouse

Redshift is AWS's warehouse for running large SQL queries over huge datasets, just like BigQuery.

ANALYTICS · AZURE EQUIVALENT

Azure Synapse Analytics

GOOGLE CLOUD

BigQuery

≈

MICROSOFT AZURE

Azure Synapse

Analytics

Synapse brings big-data SQL and dashboards together, mirroring the role BigQuery plays on Google Cloud.

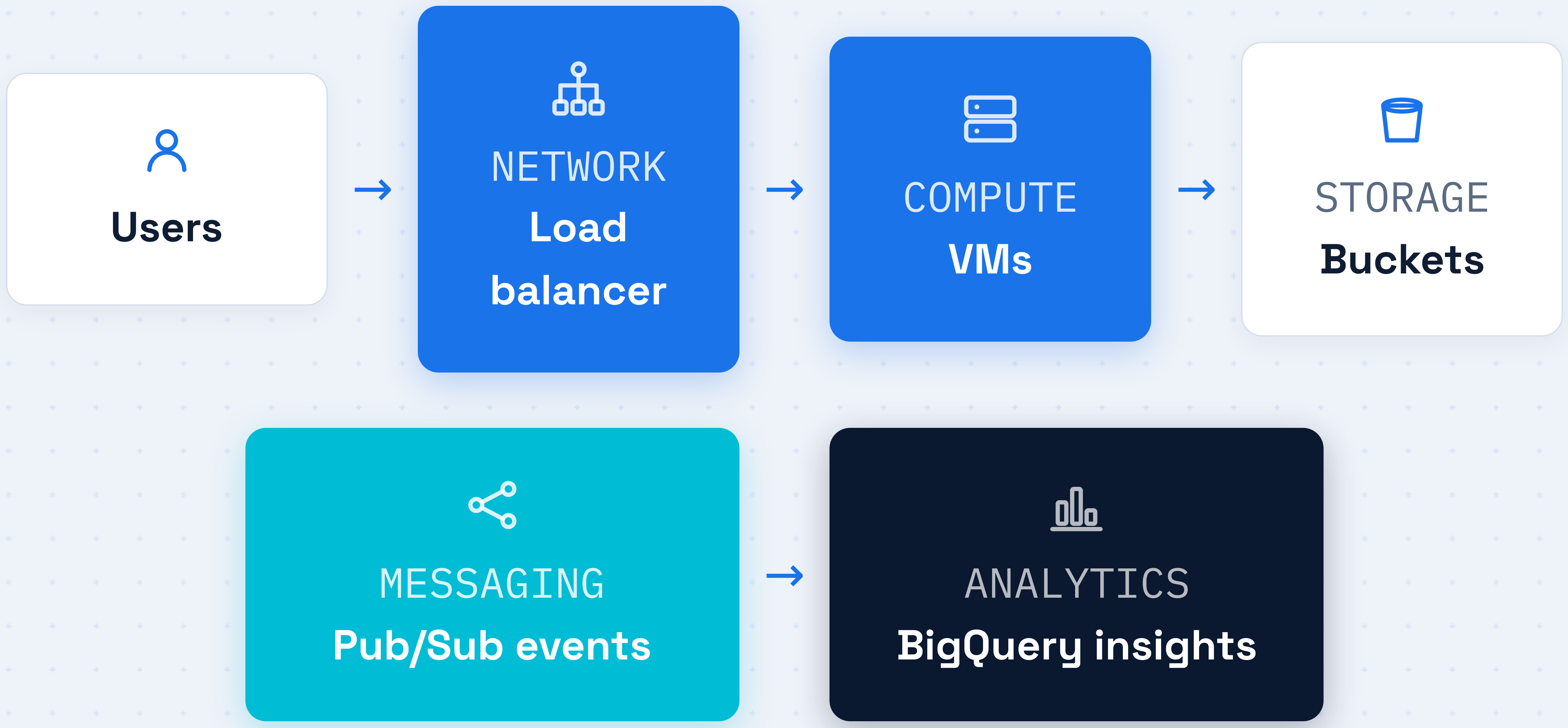
SECTION 06

Putting It Together

How the five service families combine in one real system

ARCHITECTURE

A Complete Startup System



All five families work together: traffic flows in, data is stored, events fan out, and analytics close the loop.

RECAP

The Five Service Families

01

Compute

VMs that run your code

02

Networking

Safe paths for traffic

03

Storage

Where data lives

04

Messaging

Services talk freely

05

Analytics

Data becomes answers

MIND MAP

Cloud Services at a Glance

CORE
Cloud Services →

- **Compute** — Engine · EC2 · Azure VM
- **Networking** — VPC · Firewalls · LB
- **Storage** — Object · Block · File
- **Messaging** — Pub/Sub · SNS · SQS
- **Analytics** — BigQuery · Redshift · Synapse

WRAP UP

Now We Speak Cloud

Compute, networking, storage, messaging, and analytics are the building blocks behind almost every modern application.

Google Cloud

AWS

Azure

Lecture by **Imran Sayyed**